Midterm Assessment Question Responses ITAI2376

1 Understanding Diffusion

1.1 Explain the forward diffusion process and reference visualization examples

During diffusion, random Gaussian noise is gradually added to the image; the longer the process continues, the more details disappear until only random noise remains. Because the added noise is known during training, the model can predict it from the noisy image and the timestep.

For example, a handwritten digit has a clear shape. When the noise level is low, its outline is still clear. If the noise level is moderate, then the outline becomes harder to distinguish. However, when the noise level is high, the digit is almost completely hidden.

In the visualization, it is necessary to include the figure number and the clean, moderately noisy, and heavily noisy images should be described.

1.2 Why noise is added gradually and how it affects learning

By gradually increasing the amount of noise, we obtain a range of feasible denoising tasks. The model learns across a range of noise levels, from minor disturbances to those that severely corrupt the images. This benefits generation because the inverse process can produce an image through a series of small adjustments.

Even though the process is described as sequential, you can still generate an image at a specific timestep during training. that case, you don't need to run the noise-adding steps for each training example.

1.3 When images become recognizable during denoising

The process starts with random noise; in the initial stages, the overall structure forms, while in the later stages the outline and details are refined. When the image becomes recognizable depends on the noise schedule and how extensively the model has been trained.

2 Model Architecture

2.1 Why U-Net is well suited for diffusion models

U-Net consists of an encoder and a decoder; the encoder learns to recognize patterns by reducing the spatial resolution, Tion and the decoder then increases the resolution to generate a prediction with the same spatial dimensions as the input.

Diffusion models can use this type of structure since the prediction of noise requires not only a general understanding of the image but also exact information regarding individual locations; for instance, in the case of a handwritten digit, the model has to identify the overall shape while at the same time keeping the positions of its strokes.

2.2 Purpose and importance of skip connections

Skip connections pass features from the encoder directly to the corresponding decoder layers, helping restore spatial information that would otherwise be lost due to down sampling.

The decoder can combine these features overall, so the training conditioning works

The model is given a label which indicates the category instance; in the case of MNIST, the label would be one of the digits from 0 to 9.

The label is turned into a numerical form, for example, as a one-hot vector or as a learned embedding; the conditioning network then transforms this representation and combines it with the intermediate U-Net features, the way in which the combination is carried out depending on the specific implementation and possibly involving addition, multiplication, or concatenation.

To predict noise, the model takes three inputs: the noisy image, the timestep, and the requested class. When generating an image, assigning a particular class label causes the denoising process to target that category.

3 Training Analysis

3.1 The meaning of the loss value

The mean squared error, since the model predicts noise, measures the difference between the predicted noise and the noise actually added to the image; in most cases, a smaller loss indicates that the model has learned to predict this under the training conditions.

A loss does not indicate how realistic an image is; furthermore, generated samples must be examined for recognizability, diversity, and undesirable artifacts. The validation loss assesses whether the model can handle examples outside its training data.

3.2 How generated image quality changed throughout training

The samples produced by the model before it has learned the dataset are expected to be poorly structured, but as training goes on, the images should become more identifiable in shape and have sharper details.

Comparisons are most useful when you use the same initial noise and class labels at each checkpoint, as this removes variation not due to training progress.

3.3 Purpose of time embedding

The U-Net determines which diffusion timestep it is processing through time embedding; since different timesteps correspond to different noise levels, the model needs this information to interpret the noisy image correctly.

The timestep is turned into a vector and included in the network's features, enabling adjustments to its predictions throughout the noise schedule; at the end, images are highly corrupt.

4 CLIP Evaluation

4.1 The information that the CLIP scores provide about generated images

CLIP represents both images and text descriptions as vectors; the closer they are, the more similar they are, showing how well the image matches a description like "a photograph of an airplane".

The higher the similarity score, the greater the alignment between the image and the text, but this does not imply the image is realistic, original, or free of defects. Moreover, a raw CLIP similarity score is not a percentage probability that a classification is correct.

CLIP may be less informative for small grayscale handwritten digits than for natural images; therefore, consider its results alongside visual inspection and, where appropriate, a classifier trained on the selected dataset.

4.2 Why certain images may be easier or harder to generate

I think it is simpler to learn categories with regular shapes and less variation, since categories with complicated structures and a wide range of viewpoints, backgrounds, or fine details require the model to consider more possibilities.

Regarding MNIST, a simple digit is easier to generate clearly than one with several curves or loops; for CIFAR-10, differences in pose, lighting, and background can make generation more difficult. These hypotheses should be tested with actual examples.

4.3 How CLIP scores could improve generation

The generated candidates could be ranked according to the requested class description, and categories that require further investigation could be identified.

Another possibility is to use CLIP guidance, since aligning the image with the text influences the sampling process. The drawback is that it requires additional effort and computational power.

Focusing solely on CLIP similarity might produce images that match a description but contain visual defects; CLIP should therefore be used alongside other quality checks.

5 Practical Applications

5.1 Potential real-world applications

Diffusion models can generate images for a variety of purposes, such as design, education, entertainment, and product visualization. Diffusion systems like these can also perform image editing, inpainting, and restoration, provided they are trained or set up for those tasks.

Another application is creating synthetic training data; for example, a digit-generation model could generate additional examples of handwriting recognition experiments. The images we provide are designed to offer variation and avoid misleading patterns.

5.2 Limitations of the current model

The dataset size, image resolution, network capacity, and available training time all limit the model's capabilities; a model trained exclusively on MNIST learns to recognize handwritten digits, not general scenes or objects.

Because many denoising steps are required, this process can be computationally expensive. The samples produced might have distorted shapes, an unclear class identity, or a lack of diversity. Relying entirely on the loss function to predict noise is not enough to ensure high-quality output samples.

The principal limitations identified in our model were [describe the problems supported by your generated examples].

5.3 The three specific improvements and the reasons why they would be helpful

To achieve this, you need to train the system longer while closely monitoring the validation results and the fixed sample images. It is a good idea to save checkpoint files somewhere where the output.

2. While performing sampling, you should use an exponential moving average of the model weights since averaging the weights over the training updates has the effect of reducing the impact of short-term fluctuations and thus results in more stable samples; it is necessary to compare the advantages of this method with those of the original model.

3. The classifier-free guidance should be implemented by training the model while omitting some of the class labels, in this way enabling it to generate both conditional and unconditional predictions; during the sampling process, these predictions should be combined in order to enhance class alignment. Since too much guidance might reduce diversity or cause artifacts, test a range of guidance strengths.